Code No: R31042 **R10** **Set No: 1**

III B.Tech. I Semester Regular and Supplementary Examinations, December-2013

DIGITAL IC APPLICATIONS

(Comm to ECE, EIE, BME, ECC)

Time: 3 Hours **Max Marks: 75**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Design CMOS NOR gate and analyze its behavior using switch models.
   b) Design and explain 2:1 multiplexer using CMOS transmission gates.

2. a) Distinguish between different logic families with respect to various performance parameters.
   b) Design ECL OR/NOR gate and explain its working.

3. a) Design and explain a combinational logic circuit whose output is five times of given three bit input.
   b) A combinational logic circuit is defined by the following Boolean functions.
   F1 = (ABC)' + A'C
   F2 = A(BC)' + A'B
   F3 = AB'C + AB'
   Design the circuit with a decoder.

4. a) Design and explain a BCD to Seven segment display decoder.
   b) Design and explain four bit parallel adder.

5. a) Design and explain SR latch using universal logic gates.
   b) Realize JK-FF using D-FF. Draw the necessary logic diagrams and explain the operation.

6. a) Explain the design procedure and working principle of Johnson counter.
   b) Design, draw and explain a four bit serial adder.

7. Implement the following Boolean functions using PAL.
   F1(A,B,C,D)= ∑m (0,1,3,5,6,7,8,12)
   F2(A,B,C,D)=∑m (1,3,4,5,6,7,8,10,11,12)
   F3(A,B,C,D)=∑m (0,1,2,3,4,5,6,9,11,15)
   Draw the necessary logic diagrams and explain.

8. a) Write a short note on commercial ROM types.
   b) Construct and explain 128x8 ROM using 32x8 ROMs.

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Code No: R31042 **R10** **Set No: 2**

III B.Tech. I Semester Regular and Supplementary Examinations, December-2013

DIGITAL IC APPLICATIONS

(Comm to ECE, EIE, BME, ECC)

Time: 3 Hours **Max Marks: 75**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Design CMOS NAND gate and analyze its behavior using switch models.
   b) Discuss the dynamic electrical behavior of CMOS logic circuits.

2. a) Distinguish between different logic families with respect to Voltage, Power and Delay parameters.
   b) Design a CMOS circuit for the logical expression F= (A.(B’+C))’.

3. a) Design a combinational circuit whose output is square of the given three bit input.
   b) Implement given logical expression using 8:1 multiplexer.
   $F=\sum m(0,2,6,7,8,9,10,12,14)$

4. a) Implement 4x16 decoder using IC 74LS139.
   b) Design Eight bit ripple carry adder and explain its operation.

5. a) Realize D-FF and T-FF using JK-FF. Explain with logic diagrams and functional tables.
   b) Explain synchronous and ripple counters. Compare their merits and demerits.

6. Design, draw and explain the working of Universal shift register.

7. a) Implement the following Boolean function using PLA.
   $F1(A,B,C,D)= \sum M (0,1,2,5,6,7,8,12)$
   $F2(A,B,C,D)=\sum M (3,5,6,7,8,10,11,12)$
   $F3(A,B,C,D)=\sum M (0,3,4,5,6,9,11,15)$
   b) Differentiate PLA and PAL with different performance parameters.

8. a) Draw and explain SRAM internal construction.
   b) Explain Read and Write operations in DRAM with relevant timing diagrams.

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Code No: R31042 **R10** **Set No: 3**

III B.Tech. I Semester Regular and Supplementary Examinations, December-2013

DIGITAL IC APPLICATIONS

(Comm to ECE, EIE, BME, ECC)

Time: 3 Hours **Max Marks: 75**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Design CMOS NAND gate and analyze its behavior using switch models.
   b) Discuss the steady state electrical behavior of CMOS circuits.

2. a) Illustrate CML logic with the help of inverter/buffer circuit.
   b) What is the need of interfacing different logic families and explain how to perform CMOS/TTL interfacing.

3. a) Using 4:1 multiplexer design a Car buzzer which blows when seat belt is not properly fit or when door is opened or when hand brakes are applied. All it should happen only when the ignition is ON.
   b) Implement an inverter using multiplexer and explain its operation.

4. a) Design, draw and explain the working of barrel shifter.
   b) Design, draw and explain look ahead carry generator.

5. What is meant by race around condition? How this can be eliminated in master slave JK-flip-flop. Explain with suitable diagrams.

6. a) Explain the design procedure for an asynchronous Mod 10 counter.
   b) Explain the design and working of parallel-in serial-out shift register.

7. a) Implement the following Boolean function using PLA.
   $$F1\ (A, B, C) = \sum m\ (0, 1, 3, 7)$$
   $$F2\ (A, B, C) = \sum m\ (1, 2, 4, 5, 6)$$
   b) Implement full subtractor using PROM. Explain.

8. a) Design, draw and explain 128x8 ROM using 32x8 ROMs.
   b) Explain Read and Write operations in DRAM.

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Code No: R31042 **R10** **Set No: 4**

III B.Tech. I Semester Regular and Supplementary Examinations, December-2013

DIGITAL IC APPLICATIONS

(Comm to ECE, EIE, BME, ECC)

Time: 3 Hours **Max Marks: 75**

Answer any FIVE Questions

All Questions carry equal marks

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1. a) Design static CMOS inverter and analyze its behavior using switch models.
   b) Discuss the pros and cons of large and small pull-up resistance.

2. a) List out the characteristics of ECL family. Explain, in brief.
   b) Design, draw and explain CMOS circuit for the logical expression F= (A+(B'.C))'.

3. a) Implement 4x16 decoder using ICs 74LS138 and other logic gates.
   b) Implement given logical expression using 8:1 multiplexer.
   $F=\sum m(0,2,6,7,8,9,10,12,14)$

4. a) What is dual priority encoder? Explain.
   b) Explain the design and working of IC 74LS85, 4-bit comparator.

5. Draw the circuit diagram of positive edge triggered J-K flip-flop using NOR gates and explain its operation using truth table. How race around conditions are eliminated?

6. a) Design a 4-bit ring counter using D- flip flops. Draw and explain the circuit diagram and timing diagrams.
   b) Realize D-FF using T-FF and explain the operation.

7. a) Implement Full adder using PROM and explain.
   b) Implement the following Boolean function using PLA.
   $F1\ (A,B,C)= \sum m\ (0,1,3,5,)$
   $F2\ (A,B,C)=\sum m\ (1,3,4,5,6)$

8. a) Differentiate Static RAM and Dynamic RAM with various parameters.
   b) Write a short note on commercial ROM types.

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